

Chemistry 3A

Introductory General Chemistry

ACS Final Review

ABBREVIATIONS AND SYMBOLS

amount of substance	n	Faraday constant	F	molar mass	M
ampere	A	free energy	G	mole	mol
atmosphere	atm	frequency	ν	Planck's constant	h
atomic mass unit	u	gas constant	R	pressure	P
Avogadro constant	N_A	gram	g	rate constant	k
Celsius temperature	$^{\circ}\text{C}$	hour	h	reaction quotient	Q
centi- prefix	c	joule	J	second	s
coulomb	C	kelvin	K	speed of light	c
density	d	kilo- prefix	k	temperature, K	T
electromotive force	E	liter	L	time	t
energy of activation	E_a	measure of pressure mm Hg		vapor pressure	VP
enthalpy	H	milli- prefix	m	volt	V
entropy	S	molal	m	volume	V
equilibrium constant	K	molar	M		

CONSTANTS

$$R = 8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$$

$$R = 0.0821 \text{ L} \cdot \text{atm} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$$

$$1 F = 96,500 \text{ C} \cdot \text{mol}^{-1}$$

$$1 F = 96,500 \text{ J} \cdot \text{V}^{-1} \cdot \text{mol}^{-1}$$

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

$$h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$$

$$c = 2.998 \times 10^8 \text{ m} \cdot \text{s}^{-1}$$

$$0^{\circ}\text{C} = 273.15 \text{ K}$$

PERIODIC TABLE OF THE ELEMENTS

1 1A	2 2A											13 3A	14 4A	15 5A	16 6A	17 7A	18 8A
1 H 1.008																	2 He 4.003
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (277)	113 (Uut)	114 (Uuq)	115 (Uup)	116 (Uuh)	117 (Uus)	118 (Uuo)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

1. Add 7.77 g to 11.666 g and round the answer to the appropriate place.
A) 19.43 g B) 19.436 g
C) 19.44 g D) 19.0 g

2. How many significant digits are in the length measurement 10.50 cm?
A) 1 B) 2 C) 3 D) 4

3. A block has a mass of 124.7 g with a volume of 25.0 cm^3 . What is the block's density?
A) 0.498 g/cm^3 B) 4.99 g/cm^3
C) 3117.5 g/cm^3 D) 50 g/cm^3

4. What quantity is expressed by the metric unit kilogram?
A) time B) mass
C) volume D) length

5. The density of ether is 0.714 g/mL. What is the mass of 10.0 mL of ether?

- A) 0.0714 g B) 71.4 g
C) 7.14 g D) 1.40 g

6. Liquid xenon boils at 166 K. What is the boiling point on the Celsius scale?

- A) 439 °C B) -439 °C
C) 107°C D) -107°C

7. Refer to a periodic table and predict which of the following is a metal.

- A) Sb B) Ge
C) Na D) all of the above

8. Using atomic notation, indicate the isotope having 11 p⁺, 12 n⁰, and 11 e⁻.

- A) $^{12}_{11}\text{Mg}$ B) $^{12}_{11}\text{Na}$
C) $^{23}_{12}\text{Na}$ D) $^{23}_{11}\text{Na}$

9. Lithium occurs naturally as ${}^6\text{Li}$ and ${}^7\text{Li}$. Which isotope is more abundant? (Hint: Refer to the Periodic Table.)

A) lithium-7

B) lithium-6

C) lithium-3

D) lithium-4

10. Which of the following elements has the largest atomic radius?

A) Li

B) Na

C) Mg

D) H

11. Which of the following has chemical properties most similar to zinc?

A) Ga

B) Ag

C) Cd

D) Cu

12. What is the core notation for the electron configuration of an aluminum atom?

A) $[\text{Ne}] 3s^2 3p^1$

B) $[\text{Ne}] 2s^2 2p^1$

C) $[\text{Ne}]$

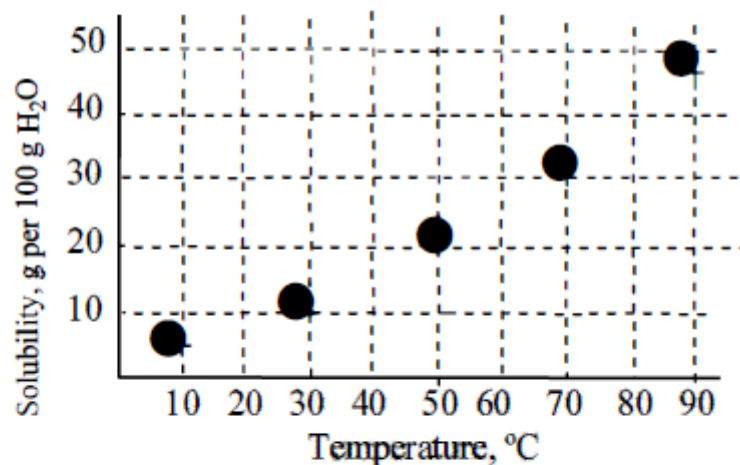
D) $[\text{Ne}] 3s^2 3d^1$

13. Predict the number of valence electrons for a Group IIA/2 element.
- A) 3 B) 1 C) 2 D) 8
14. The NH_4^+ ion is classified as which of the following?
- A) monoatomic cation
B) monoatomic anion
C) polyatomic cation
D) polyatomic anion
15. How many atoms of copper are in 2.50 mol Cu metal?
- A) 2.41×10^{24} atoms
B) 2.41×10^{23} atoms
C) 1.51×10^{24} atoms
D) 1.51×10^{22} atoms
16. What is the chemical formula for cobalt(II) bromide?
- A) Co_2Br B) Co_2Br_3
C) Co_3Br_2 D) CoBr_2

17. Which of the following is the strongest intermolecular force between water molecules?
- A) dipole dipole forces
 - B) hydrogen bond
 - C) dispersion forces
 - D) polar covalent bond
18. What is the mass of 0.0747 moles of gold, Au?
- A) 0.0679 g
 - B) 13.3 g
 - C) 0.0748 g
 - D) 14.7 g
19. If the pressure of 50.0 mL of oxygen gas increases from 735 mm Hg to 925 mm Hg, what is the final volume? (Assume temperature remains constant.)
- A) 50.0 mL
 - B) 62.9 mL
 - C) 39.7 mL
 - D) 48.4 mL
20. What is the chemical formula for the acetate ion?
- A) HCO_3^{2-}
 - B) $\text{C}_2\text{H}_3\text{O}_2^-$
 - C) HCO_3^-
 - D) $\text{C}_2\text{H}_3\text{O}_2^{2-}$

- | | |
|-------|-------|
| 1. C | 11. C |
| 2. D | 12. A |
| 3. B | 13. C |
| 4. B | 14. C |
| 5. C | 15. C |
| 6. D | 16. D |
| 7. C | 17. B |
| 8. D | 18. D |
| 9. A | 19. C |
| 10. B | 20. B |

1. A student is asked to precisely dispense 24.70 mL of a solution. Which item glassware could be used for this task?
(A) 50 mL beaker
(B) 50 mL buret
(C) 50 mL Erlenmeyer flask
(D) 50 mL graduated cylinder
2. The solubility of KClO_3 at several temperatures is shown in the accompanying diagram.



If a student mixes 10.0 g of KClO_3 with 45.0 g of H_2O at 70°C . Which statement about the final solution is correct?

- (A) It is a saturated solution.
- (B) It is a supersaturated solution.
- (C) It is an unsaturated solution.
- (D) It is impossible to determine.

3. Cigarette lighters are generally filled with butane. When butane burns in air, heat is given off to the surroundings. Which of the following is true?
- (A) The reaction is endothermic.
 - (B) The reaction is exothermic.
 - (C) The reaction is a precipitation reaction.
 - (D) The reaction is an acid-base reaction.
-
4. The phase transition from solid to liquid is called
- (A) condensation.
 - (B) evaporation.
 - (C) melting/fusion.
 - (D) sublimation.
5. In the Lewis structure for NH_3 , how many lone pairs of electrons are there?
- (A) 0
 - (B) 1
 - (C) 2
 - (D) 4
6. What is the concentration of nitric acid in a solution made by diluting 100 mL of a 0.200 M HNO_3 solution to a final volume of 200 mL?
- (A) 0.133 M
 - (B) 0.150 M
 - (C) 0.167 M
 - (D) 0.100 M

7. Typical grocery store bags are made of polyethylene, which are linked molecules of ethylene (C_2H_4). If a bag weighs 12.4 g. How many molecules of ethylene (C_2H_4) are in this bag?

- (A) 1.36×10^{24} (B) 6.02×10^{23}
(C) 5.33×10^{23} (D) 2.67×10^{23}

8. A sulfide ion (S^{2-}) contains?

- (A) 16 electrons (B) 32 protons
(C) 32 neutrons (D) 18 electrons

9. What is the oxidation number of Chromium in $Cr(SO_4)_2$?

- (A) +1 (B) +2 (C) +4 (D) +6

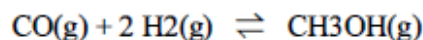
10. How many Liters is 122 mL?

- (A) 0.122 L (B) 1.22×10^5 L
(C) 1.22 L (D) 1.22×10^{-3} L

11. How many protons and neutrons are in ^{59}Ni ?

- (A) 59 protons and 47 neutrons
- (B) 28 protons and 59 neutrons
- (C) 59 protons and 28 neutrons
- (D) 28 protons and 31 neutrons

12. For the following reaction at equilibrium, addition of CO(g) would cause ____.

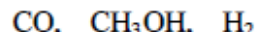


- (A) a shift left (more reactants formed).
- (B) a shift right (more products formed).
- (C) No change.
- (D) The reaction to slow down.

13. What volume of a 1.50 M KOH solution would contain 0.750 moles of potassium?

- (A) 2 L
- (B) 0.250 L
- (C) 500 mL
- (D) 1 gal

14. Put the following in order of increasing strength of intermolecular forces



- (A) $\text{CO} < \text{CH}_3\text{OH} < \text{H}_2$
- (B) $\text{CH}_3\text{OH} < \text{H}_2 < \text{CO}$
- (C) $\text{H}_2 < \text{CO} < \text{CH}_3\text{OH}$
- (D) $\text{CO} < \text{H}_2 < \text{CH}_3\text{OH}$

15. What is the molecular shape of phosphorus in PH_3 ?

- (A) tetrahedral
- (B) trigonal planar
- (C) trigonal pyramidal
- (D) linear

16. What is the correct formula for Iron (III) Oxide?

- (A) Fe_3O
- (B) Fe_2O_3
- (C) Fe_3O_2
- (D) FeO

17. Which of the following is the ground state electron configuration of a magnesium atom?

- (A) $1s^2 2s^2 2p^6$
- (B) $1s^2 2s^2 2p^6 3s^2$
- (C) $2p^6 3s^1$
- (D) $1s^2 2s^2 2p^6 3s^2 4s^1$

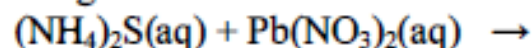
18. A sample of oxygen gas has a pressure of 1.05 atm and a volume of 22 L. What pressure would be required to compress the gas to 9.0 L?

- (A) 2.6 atm
- (B) 11.6 atm
- (C) 0.43 atm
- (D) 297 torr

19. How many significant digits are in 0.0219 g ?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

20. What insoluble product would be formed by the following reaction?



- (A) N_2H_8 (s)
- (B) $(\text{NH}_4)_2(\text{NO}_3)_2$ (s)
- (C) PbS (s)
- (D) All products are soluble

Answer Key

1)B 2)C 3)B 4)C 5)B 6)D 7)D 8)D 9)C 10)A

11)D 12)B 13)C 14)C 15)C 16)B 17)B 18)A 19)B 20)C
